APPLE ITUNES MUSIC ANALYSIS

SECTION 1 — DATA OVERVIEW & STRUCTURE

Observation:

The database consists of 11 relational tables including customers, invoices, invoice lines, tracks, albums, artists, genres, employees, playlists, and media types.

Key Structural Insights:

• Proper normalization with foreign key relationships.
• Invoice → Invoice_Line drives revenue logic.
• Track → Album → Artist forms content hierarchy.
• Customer → Support_Rep connects operational layer.
• Genre and Media_Type allow product segmentation.

Interpretation:

The schema supports multi-dimensional analysis:

- Revenue
- Customer behavior
- Product performance
- Geographic trends
- Operational efficiency

SECTION 2 — SALES & REVENUE ANALYSIS

1. <u>Monthly Revenue Trends (Last Two Years)</u>

```sql
SELECT DATE_TRUNC('month', invoice_date) AS month,ROUND(SUM(total), 2) AS monthly_revenue
FROM invoice
WHERE invoice_date >= (SELECT MAX(invoice_date) FROM invoice) - INTERVAL '2 years'
GROUP BY month
ORDER BY month;
```

Output :-

Data Output Messages Notifications

	month timestamp without time zone	monthly_revenue numeric
1	2019-01-01 00:00:00	85.14
2	2019-02-01 00:00:00	59.40
3	2019-03-01 00:00:00	125.73
4	2019-04-01 00:00:00	116.82
5	2019-05-01 00:00:00	104.94
6	2019-06-01 00:00:00	92.07
7	2019-07-01 00:00:00	124.74
8	2019-08-01 00:00:00	159.39
9	2019-09-01 00:00:00	126.72
10	2019-10-01 00:00:00	43.56
11	2019-11-01 00:00:00	71.28
12	2019-12-01 00:00:00	111.87
13	2020-01-01 00:00:00	43.56
14	2020-02-01 00:00:00	97.02
15	2020-03-01 00:00:00	78.21
16	2020-04-01 00:00:00	119.79
17	2020-05-01 00:00:00	82.17
18	2020-06-01 00:00:00	120.78
19	2020-07-01 00:00:00	76.23
20	2020-08-01 00:00:00	72.27
21	2020-09-01 00:00:00	95.04
22	2020-10-01 00:00:00	144.54
23	2020-11-01 00:00:00	83.16
24	2020-12-01 00:00:00	125.73

Observation:
Revenue fluctuates month-to-month without a strong upward growth trend.

Interpretation:
- No aggressive growth trajectory.
- Business appears stable but mature.
- Likely dependent on repeat buyers rather than acquisition spikes.

2. Average Invoice Value

```sql
/* 2.2. What is the average value of an invoice (purchase)? */

SELECT ROUND(AVG(total), 2) AS avg_invoice_value
FROM invoice;
```

Output :-

Data Output | Messages | Notifications

	avg_invoice_value numeric
1	7.67

Average invoice value = 7.67

Interpretation:
- Customers purchase small baskets.
- Micro-transaction driven model.
- Revenue depends on frequency, not ticket size.

3. Revenue Contribution by Sales Representatives

```sql
/* 2.4. How much revenue does each sales representative contribute? */

SELECT e.employee_id,e.first_name || ' ' || e.last_name AS employee_name,ROUND(SUM(i.total), 2) AS revenue_generated
FROM employee e
JOIN customer c
ON e.employee_id = c.support_rep_id
JOIN invoice i
ON c.customer_id = i.customer_id
GROUP BY e.employee_id, employee_name
ORDER BY revenue_generated DESC;
```

Output :-

	employee_id [PK] integer	employee_name text	revenue_generated numeric
1	3	Jane Peacock	1731.51
2	4	Margaret Park	1584.00
3	5	Steve Johnson	1393.92

Observation:

Revenue distribution:
1. Jane Peacock → 1731.51
2. Margaret Park → 1584.00
3. Steve Johnson → 1393.92

Interpretation:
- Revenue is relatively balanced.
- No extreme employee dependency.
- Operational risk low.

4. Peak Months & Quarters

```
/* 2.5. Which months or quarters have peak music sales? */

/* Monthly peak */
SELECT DATE_PART('month', invoice_date) AS month_number,ROUND(SUM(total), 2) AS revenue
FROM invoice
GROUP BY month_number
ORDER BY revenue DESC;
```

Output:-

	month_number double precision	revenue numeric
1	3	456.39
2	4	442.53
3	1	438.57
4	8	426.69
5	2	414.81
6	7	395.01
7	9	386.10
8	6	380.16
9	5	368.28
10	12	364.32
11	10	345.51
12	11	291.06

```
/* Quarterly peak */
SELECT DATE_PART('quarter', invoice_date) AS quarter,ROUND(SUM(total), 2) AS revenue
FROM invoice
GROUP BY quarter
ORDER BY revenue DESC;
```

Output:-

	quarter double precision	revenue numeric
1	1	1309.77
2	3	1207.80
3	2	1190.97
4	4	1000.89

Observation:

Highest revenue month cluster:
March and Q1 lead overall.

Quarter ranking:
Q1 highest → Q3 → Q2 → Q4.

Interpretation:
- Seasonal spike in early years.
- Likely post-holiday digital consumption behavior.

Strategic Insight:
Campaigns before Q1 may amplify revenue peak.

Sales & Revenue Insights:

1. Revenue shows moderate month-to-month fluctuations with no consistent upward growth trend.

2. Average invoice value (7.67) indicates micro-transaction purchasing behavior.

3. Sales representative contributions are relatively evenly distributed, reducing operational concentration risk.

4. Q1 consistently generates the highest revenue, while Q4 is the weakest quarter.

5. March is the strongest revenue month, suggesting early-year purchasing momentum.

SECTION 3 — PRODUCT & CONTENT ANALYSIS

1. Top Revenue-Generating Tracks

```
/* 3.1. Which tracks generated the most revenue? */
SELECT t.track_id,t.name AS track_name,ROUND(SUM(il.unit_price * il.quantity), 2) AS track_revenue
FROM track t
JOIN invoice_line il
ON t.track_id = il.track_id
GROUP BY t.track_id, track_name
ORDER BY track_revenue DESC
LIMIT 10;
```

Output:-

	track_id [PK] integer	track_name character varying (255)	track_revenue numeric
1	3336	War Pigs	30.69
2	1495	Highway Chile	13.86
3	1489	Are You Experienced?	13.86
4	1490	Hey Joe	12.87
5	1487	Third Stone From The S...	12.87
6	6	Put The Finger On You	12.87
7	1483	Love Or Confusion	11.88
8	2558	Radio/Video	11.88
9	1493	51st Anniversary	10.89
10	1129	Dead And Broken	10.89

Observations:

- Massive Outlier: "War Pigs" ($30.69) generates more than double the revenue of the runner-up.
- Sequential Clustering: Six of the top ten tracks (IDs 1483–1495) are tightly grouped, indicating one powerhouse album.
- Revenue Plateaus: Multiple tracks are tied at the exact same revenue levels (e.g., $13.86, $12.87), showing identical purchase volumes.

Interpretation:

- Album-Centric Buying: High ranking is driven by album loyalty rather than isolated "hit" singles.
- Long-Tail Model: Even the #1 track has relatively low absolute sales (approx. 31 units), confirming a niche-driven business.

Strategic Insight:

- Cross-Sell Priority: Marketing should focus on "Complete the Album" bundles to capitalize on existing grouping behavior.

2.Most Frequently Purchased Albums

```sql
/* 3.2. Which albums are most frequently included in purchases? */
SELECT a.album_id,a.title AS album_title,
COUNT(il.invoice_line_id) AS times_purchased
FROM album a
JOIN track t
ON a.album_id = t.album_id
JOIN invoice_line il
ON t.track_id = il.track_id
GROUP BY a.album_id, album_title
ORDER BY times_purchased DESC
LIMIT 10;
```

Output:-

	album_id [PK] integer	album_title character varying (255)	times_purchased bigint
1	120	Are You Experienced?	187
2	88	Faceless	96
3	207	Mezmerize	93
4	119	Get Born	90
5	214	The Doors	83
6	215	The Police Greatest Hits	80
7	185	Greatest Hits I	80
8	5	Big Ones	80
9	163	From The Muddy Banks Of The Wishkah [li...	78
10	221	My Generation - The Very Best Of The Who	76

Albums like:
- “Are You Experienced?”
- “Faceless”
- “Mezmerize”

Observation:
High album-level clustering.

Interpretation:
Customers often purchase multiple tracks from the same album.

3. Tracks Never Purchased

```
/* 3.4. Are there any tracks or albums that have never been purchased? */
SELECT t.track_id,t.name
FROM track t
LEFT JOIN invoice_line il
ON t.track_id = il.track_id
WHERE il.track_id IS NULL;
```

Output:-

	track_id [PK] integer	name character varying (255)
1	2848	Six Months Ago
2	251	Um Passeio No Mundo Livre
3	2026	Às Vezes
4	3028	Zooropa
5	2024	Deus
6	264	Amor De Muito
7	3404	Miserere mei, Deus
8	2847	Homecoming
9	1350	Powerslave
10	1070	16 Toneladas
11	2311	Low
12	276	Maracatu De Tiro Certeiro
13	1083	O Amor Daqui De Casa

Observation:

- 1697 tracks have zero purchases.

Critical Insight:

- Large long-tail inventory inefficiency.

This is catalog bloat.
Inventory breadth ≠ revenue contribution.

4. No. of tracks that are not purchased

```
SELECT COUNT(*)
FROM track t
LEFT JOIN invoice_line il
    ON t.track_id = il.track_id
WHERE il.track_id IS NULL;
```

Output:-

	count bigint
1	1697

Key Observations:

- **Dead Inventory:** 1,697 tracks (a huge portion of the catalog) have never been sold.
- **Resource Drain:** A large segment of the digital library is sitting idle without contributing revenue.

Interpretation & Insight:

- **Catalog Inefficiency**: The store carries high "digital dust" from underperforming genres like Latin or TV Shows.
- **Strategic Move**: Bundle these "hidden" tracks with hits or run a "Discovery" sale to clear stagnant inventory.

5. Revenue by Genre

```
/* Is revenue concentrated in a few genres or evenly spread? */
SELECT g.name AS genre,ROUND(SUM(il.unit_price * il.quantity), 2) AS genre_revenue
FROM genre g
JOIN track t
ON g.genre_id = t.genre_id
JOIN invoice_line il
ON t.track_id = il.track_id
GROUP BY g.name
ORDER BY genre_revenue DESC;
```

Output:-

	genre character varying (120)	genre_revenue numeric
1	Rock	2608.65
2	Metal	612.81
3	Alternative & Punk	487.08
4	Latin	165.33
5	R&B/Soul	157.41
6	Blues	122.76
7	Jazz	119.79
8	Alternative	115.83

```
SELECT g.name AS genre,ROUND(SUM(il.unit_price * il.quantity), 2) AS total_revenue,
ROUND(SUM(il.unit_price * il.quantity) * 100.0 / SUM(SUM(il.unit_price * il.quantity)) OVER (),2) AS revenue_percent
FROM genre g
JOIN track t ON g.genre_id = t.genre_id
JOIN invoice_line il ON t.track_id = il.track_id
GROUP BY g.name
ORDER BY revenue_percent DESC;
```

	genre character varying (120)	total_revenue numeric	revenue_percent numeric
1	Rock	2608.65	55.39
2	Metal	612.81	13.01
3	Alternative & Punk	487.08	10.34
4	Latin	165.33	3.51
5	R&B/Soul	157.41	3.34
6	Blues	122.76	2.61
7	Jazz	119.79	2.54
8	Alternative	115.83	2.46
9	Easy Listening	73.26	1.56
10	Pop	62.37	1.32

Observation:

Rock → 55.39%
Metal → 13.01%
Alternative & Punk → 10.34%

Top 3 genres ≈ 79% revenue.

Interpretation:
Extreme genre concentration.
Business is heavily dependent on Rock ecosystem.

Strategic Insight and Implications:

- Rock drives volume.
- Easy Listening and Electronic/Dance are high-yield per asset.
- Latin is severely underperforming (627 tracks → 0.26 per track).
- TV Shows and Drama are nearly dead monetization categories.

1. Revenue risk is genre-concentrated.

If Rock demand drops, revenue collapses.

2. Inventory is inefficient.

Large portions of catalog contribute almost nothing.

3. There is opportunity in high-efficiency genres like Easy Listening and Electronic/Dance.

They don't drive volume, but they monetize well per asset.

4. Underperforming genres (Latin, TV Shows, Drama) need reevaluation.

Either better marketing or reduced catalog investment.

SECTION 4 — ARTIST & GENRE PERFORMANCE

1.
2. **Top 5 Highest-Grossing Artists**

```
SELECT ar.artist_id,ar.name AS artist_name,ROUND(SUM(il.unit_price * il.quantity), 2) AS artist_revenue
FROM artist ar
JOIN album a ON ar.artist_id = a.artist_id
JOIN track t ON a.album_id = t.album_id
JOIN invoice_line il ON t.track_id = il.track_id
GROUP BY ar.artist_id, artist_name
ORDER BY artist_revenue DESC
LIMIT 5;
```

Output:-

	artist_id [PK] integer	artist_name character varying (255)	artist_revenue numeric
1	51	Queen	190.08
2	94	Jimi Hendrix	185.13
3	110	Nirvana	128.70
4	127	Red Hot Chili Peppers	128.70
5	118	Pearl Jam	127.71

All top artists belong to Rock.

Interpretation:
Genre dominance cascades to artist level.
Revenue concentration is structural.

2. Genres by Units Sold

```
SELECT g.name AS genre,SUM(il.quantity) AS tracks_sold
FROM genre g
JOIN track t ON g.genre_id = t.genre_id
JOIN invoice_line il ON t.track_id = il.track_id
GROUP BY g.name
ORDER BY tracks_sold DESC;
```

Output:-

	genre character varying (120)	tracks_sold bigint
1	Rock	2635
2	Metal	619
3	Alternative & Punk	492
4	Latin	167
5	R&B/Soul	159
6	Blues	124
7	Jazz	121
8	Alternative	117
9	Easy Listening	74
10	Pop	63
11	Electronica/Dance	55

Observation:

Rock → 2635
Metal → 619
Alternative & Punk → 492

Revenue ranking matches unit ranking.

Interpretation:
Revenue is volume-driven, not price-driven.
No premium pricing effect.

3. Top Genre per Country

```
WITH country_genre_sales AS (
SELECT c.country,g.name AS genre,SUM(il.quantity) AS tracks_sold
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
JOIN invoice_line il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
GROUP BY c.country, g.name)
SELECT * FROM (SELECT *,RANK() OVER (PARTITION BY country ORDER BY tracks_sold DESC) AS rnk
FROM country_genre_sales) sub
WHERE rnk = 1
ORDER BY tracks_sold DESC;
```

Output:-

	country character varying (100)	genre character varying (120)	tracks_sold bigint	rnk bigint
1	USA	Rock	561	1
2	Canada	Rock	333	1
3	France	Rock	211	1
4	Brazil	Rock	205	1
5	Germany	Rock	194	1
6	United Kingdom	Rock	166	1
7	Czech Republic	Rock	143	1
8	Portugal	Rock	108	1
9	India	Rock	102	1
10	Ireland	Rock	72	1

Rock dominates nearly every country.

Interpretation:

- Global uniformity in taste.
- Limited need for heavy regional genre localization.

Strategic Note:
Global Rock-focused campaigns viable.

Interpretation:

• Rock is universally preferred across markets.

• No country shows a fundamentally different taste profile.

• Localization strategy may not need drastic genre differentiation.

• Marketing can safely lean into Rock globally.

Artist & Genre Performance Insights:

1. Revenue is highly concentrated among Rock artists.

2. The top 5 artists are all Rock-based.

3. Rock dominates both revenue and units sold.

4. Rock is the top genre in nearly every country.

5. Revenue hierarchy is volume-driven rather than price-driven.

SECTION 5 — EMPLOYEE & OPERATIONAL EFFICIENCY

1. Highest-Spending Customer Portfolios

```
WITH customer_ltv AS (
SELECT c.customer_id,c.support_rep_id,SUM(i.total) AS lifetime_spending
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.customer_id, c.support_rep_id)
SELECT e.employee_id,e.first_name || ' ' || e.last_name AS employee_name,
ROUND(SUM(cl.lifetime_spending),2) AS total_managed_revenue
FROM employee e
JOIN customer_ltv cl
ON e.employee_id = cl.support_rep_id
GROUP BY e.employee_id, employee_name
ORDER BY total_managed_revenue DESC;
```

Output:-

	employee_id [PK] integer	employee_name text	total_managed_revenue numeric
1	3	Jane Peacock	1731.51
2	4	Margaret Park	1584.00
3	5	Steve Johnson	1393.92

Observation:

Revenue Managed per Employee

Jane Peacock → 1731.51

Margaret Park → 1584.00

Steve Johnson → 1393.92

Observation:

- Revenue distribution across employees is relatively balanced.

- Difference between top and bottom ≈ 337.59.

Insight:

- No single rep is disproportionately carrying the business.

- Revenue distribution appears operationally stable.

- Customer allocation is reasonably balanced.

2. Average Customers per Employee

```
/*  5.2 What is the average number of customers per employee? */
SELECT ROUND(AVG(customer_count),2) AS avg_customers_per_employee
FROM (SELECT support_rep_id,COUNT(*) AS customer_count
FROM customer
GROUP BY support_rep_id) sub;
```

Output:-

	avg_customers_per_employee numeric
1	19.67

Observation:

≈ 19.67 customers per rep.

Interpretation:
- Manageable workload.
- Capacity exists for scaling.

3. **Revenue by Employee Region**

```
/* 5.3 Which employee regions bring in the most revenue? */
SELECT e.country AS employee_country,ROUND(SUM(i.total),2) AS revenue_generated
FROM employee e
JOIN customer c ON e.employee_id = c.support_rep_id
JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY e.country
ORDER BY revenue_generated DESC;
```

Output:-

	employee_country character varying (100)	revenue_generated numeric
1	Canada	4709.43

Observation:

All employees located in Canada.
Total revenue generated centrally.

Interpretation:
Centralized support model.
Low operational fragmentation.
Potential timezone exposure risk if scaling globally.

SECTION 6 — GEOGRAPHIC TRENDS

1. Countries with Most Customers

```
/* 6.1 Which countries have the highest number of customers? */

SELECT country,COUNT(*) AS customer_count
FROM customer
GROUP BY country
ORDER BY customer_count DESC;
```

Output:-

	country character varying (100)	customer_count bigint
1	USA	13
2	Canada	8
3	France	5
4	Brazil	5
5	Germany	4
6	United Kingdom	3
7	Portugal	2
8	India	2
9	Czech Republic	2
10	Norway	1

Observation:

- The customer base is geographically diversified, but heavily skewed toward the USA.

- The USA has the largest market share of customers.

- No single country dominates overwhelmingly — this is moderate concentration.

2. Revenue by Country

```sql
/* 6.2 How does revenue vary by region? */
SELECT c.country,ROUND(SUM(i.total),2) AS total_revenue
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.country
ORDER BY total_revenue DESC;
```

Output:-

	country character varying (100)	total_revenue numeric
1	USA	1040.49
2	Canada	535.59
3	Brazil	427.68
4	France	389.07
5	Germany	334.62
6	Czech Republic	273.24
7	United Kingdom	245.52
8	Portugal	185.13
9	India	183.15
10	Ireland	114.84

Insight:
Some small markets show high spending intensity.

Strategic Insight:
USA = scale market
Czech Republic = high-value micro market

Growth lever: Improve monetization in mid-tier market.

3. Underserved Geographic Regions

```sql
/* 6.3 Are there underserved geographic regions (high users, low sales)? */
WITH country_stats AS (SELECT c.country,
COUNT(DISTINCT c.customer_id) AS customers,SUM(i.total) AS revenue
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.country)
SELECT country,customers,ROUND(revenue,2) AS revenue,ROUND(revenue / customers,2)
FROM country_stats
ORDER BY customers DESC;
```

Output:-

	country character varying (100)	customers bigint	revenue numeric	revenue_per_customer numeric
1	USA	13	1040.49	80.04
2	Canada	8	535.59	66.95
3	Brazil	5	427.68	85.54
4	France	5	389.07	77.81
5	Germany	4	334.62	83.66
6	United Kingdom	3	245.52	81.84
7	Portugal	2	185.13	92.57
8	India	2	183.15	91.58
9	Czech Republic	2	273.24	136.62
10	Italy	1	50.49	50.49
11	Netherlands	1	65.34	65.34

Observation:

The Czech Republic has only 2 customers — but extremely high spending per customer.

- This is a high-value micro-market.

- India and Portugal also punch above their weight.

- The USA has the most customers but moderate per-customer value.

This tells us:

- USA = scale market

- Czech Republic = high-intensity market

SECTION 7 — CUSTOMER RETENTION & PURCHASE PATTERNS

1. Purchase Frequency Distribution

```
/* 7.1 Distribution of purchase frequency per customer */

SELECT purchase_count,COUNT(*) AS number_of_customers
FROM (SELECT customer_id, COUNT(*) AS purchase_count
FROM invoice
GROUP BY customer_id) sub
GROUP BY purchase_count
ORDER BY purchase_count;
```

Output:-

	purchase_count bigint	number_of_customers bigint
1	4	1
2	5	1
3	7	1
4	8	6
5	9	12
6	10	12
7	11	9
8	12	8
9	13	6
10	15	1

Observation-

Most customers purchase between 8–12 times.

Interpretation:
 Strong repeat behavior.
 Not one-time buyers.

2. Average Time Between Purchases

```sql
/*  7.2 Average time between purchases */
WITH purchase_gaps AS (
SELECT customer_id,invoice_date,
LAG(invoice_date) OVER (PARTITION BY customer_id
ORDER BY invoice_date) AS prev_date
FROM invoice)
SELECT ROUND(
AVG(EXTRACT(EPOCH FROM (invoice_date - prev_date)) / 86400),2) AS avg_days_between_purchases
FROM purchase_gaps
WHERE prev_date IS NOT NULL;
```

Output:-

	avg_days_between_purchases numeric
1	132.28

Observation:

≈ 132 days (~4.4 months)

Interpretation:

- Moderate engagement cycle.
- Not high-frequency.
- Opportunity to reduce the gap.

This is the biggest growth lever.

3. Multi-Genre Purchase Percentage

```sql
/*  7.3 Percentage of customers purchasing multiple genres */
WITH customer_genre_count AS (
SELECT c.customer_id,COUNT(DISTINCT g.genre_id) AS genre_count
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
JOIN invoice_line il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
GROUP BY c.customer_id)
SELECT ROUND(COUNT(*) FILTER (WHERE genre_count > 1) * 100.0 / COUNT(*),2) AS multi_genre_percent
FROM customer_genre_count;
```

Output:-

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	multi_genre_percent numeric
1	100.00

Observation:

- 100% of customers buy multiple genres.

Critical Insight:

- Customers are not genre-locked.
- Revenue concentration is supply-driven,not customer-limitation driven.

Huge cross-sell potential already validated.

SECTION 8 — OPERATIONAL OPTIMIZATION

1. **Track Combinations Purchased Together**

```sql
/*  8.1 Most common combinations of tracks purchased together *
SELECT t1.name AS track_1,t2.name AS track_2,
COUNT(*) AS times_bought_together
FROM invoice_line il1
JOIN invoice_line il2
ON il1.invoice_id = il2.invoice_id
AND il1.track_id < il2.track_id
JOIN track t1 ON il1.track_id = t1.track_id
JOIN track t2 ON il2.track_id = t2.track_id
GROUP BY track_1, track_2
ORDER BY times_bought_together DESC
LIMIT 10;
```

Output:-

	track_1 character varying (255)	track_2 character varying (255)	times_bought_together bigint
1	Love Or Confusion	I Don't Live Today	9
2	Foxy Lady	Third Stone From The S...	9
3	May This Be Love	Highway Chile	9
4	Remember	Stone Free	8
5	May This Be Love	Purple Haze	8
6	Can You See Me	Third Stone From The S...	8
7	Can You See Me	May This Be Love	8
8	May This Be Love	Stone Free	8
9	Manic Depression	Purple Haze	8
10	I Don't Live Today	Stone Free	8

Observation-

Repeated same-artist bundling patterns.

Interpretation:
Artist loyalty is stronger than genre loyalty.
"Complete the album" suggestions are likely effective.

2. Pricing Patterns

```sql
/*  8.2 Are there pricing patterns that lead to higher or lower sales? */
SELECT t.unit_price,SUM(il.quantity) AS total_units_sold
FROM track t
JOIN invoice_line il ON t.track_id = il.track_id
GROUP BY t.unit_price
ORDER BY t.unit_price;
```

Output:-

	unit_price numeric (10,2)	total_units_sold bigint
1	0.99	4754
2	1.99	3

Observation-

0.99 → 4754 units
1.99 → 3 units

Interpretation:
- Price elasticity is extremely high.
- Higher price kills demand.

The revenue model must remain volume-based.

3. Media Type Trends

```sql
/*  8.3 Which media types are declining or increasing? */
SELECT mt.name AS media_type,DATE_TRUNC('year', i.invoice_date) AS year,
SUM(il.quantity) AS units_sold
FROM media_type mt
JOIN track t ON mt.media_type_id = t.media_type_id
JOIN invoice_line il ON t.track_id = il.track_id
JOIN invoice i ON il.invoice_id = i.invoice_id
GROUP BY media_type, year
ORDER BY media_type, year;
```

Output:-

	media_type character varying (120)	year timestamp without time zone	units_sold bigint
1	AAC audio file	2017-01-01 00:00:00	7
2	AAC audio file	2018-01-01 00:00:00	4
3	AAC audio file	2019-01-01 00:00:00	8
4	AAC audio file	2020-01-01 00:00:00	2
5	MPEG audio file	2017-01-01 00:00:00	1068
6	MPEG audio file	2018-01-01 00:00:00	1060
7	MPEG audio file	2019-01-01 00:00:00	1108
8	MPEG audio file	2020-01-01 00:00:00	1023
9	Protected AAC audio file	2017-01-01 00:00:00	126
10	Protected AAC audio file	2018-01-01 00:00:00	82

Observation-

- MPEG audio file dominant and stable.
- No major format transition observed.

Interpretation:
1. Mature product environment.
2. No technological disruption detected.

OVERALL EXECUTIVE CONCLUSION

Current State of the Business The business currently operates as a stable, mature digital music platform. The revenue model is heavily volume-driven , relying on standardized micro-transactions where demand is extremely price-sensitive. The platform benefits from a highly engaged repeat-customer base and a well-balanced operational sales structure. However, revenue is heavily structurally dependent on the Rock genre , which alone accounts for over 55% of all sales.

Primary Vulnerabilities & Risks

- **Severe Catalog Inefficiency:** The platform suffers from "catalog bloat," with 1,697 tracks—a massive portion of the inventory—generating zero lifetime purchases.
-
- **Elongated Engagement Cycles:** The average time between customer purchases is roughly 132 days (over 4 months) , leaving revenue on the table between visits.
- **Seasonal Slumps:** Quarter 4 (Q4) consistently underperforms as the weakest quarter, missing out on end-of-year consumer spending.
- **Genre Concentration Dependency:** Over-reliance on Rock limits revenue diversity; if Rock demand drops, total revenue is highly exposed.

Strategic Growth Roadmap

To transition from a passive digital library to an active, high-yield marketplace, the business should execute the following growth levers:

1. **Shorten the Retention Cycle (The #1 Growth Lever)** : Implement automated "churn prevention" and win-back campaigns triggered when a customer approaches 150 days of inactivity, directly attacking the 132-day purchase gap.
2. **Clear the "Digital Dust"** : Address the 1,697 unpurchased tracks by deploying them in low-cost "Surprise Bundles" or offering them as free promotional add-ons alongside high-performing tracks.

3. **Deploy Artist-Centric Bundling :** Because 100% of customers purchase across multiple genres and show strong artist loyalty , heavily push "Complete the Album" and dynamic "Frequently Bought Together" recommendations to increase average cart size.

4. **Capitalize on High-Intensity Micro-Markets :** While the USA is the primary scale market , the Czech Republic, India, and Portugal demonstrate incredibly high per-customer spending. Launch localized referral programs to expand these lucrative mid-tier markets.

5. **Counteract the Q4 Dip:** Introduce "Digital Gift Cards" or "Holiday Credit Bundles" in November to capture Q4 consumer spending, which will simultaneously amplify the existing early-year (Q1) revenue peak.